

Hyper-Reflectivity and Neovascularization in LMRI DB

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Q: What percent of MacTel patients have outer retina hyper reflectivity (ORHR) and choroidal neovascularization (CNV)?

ORHR and CNV are not variables in the database, but HR_TEMPORAL (Hyper reflectivity on the temporal side) and NEOVASC (Neovascularization) are.

36.6 % of eyes develop HR_TEMPORAL and **6.1%** develop NEOVASC while only **4.9%** develop both.
47.6 % of patients develop HR_TEMPORAL and **10.1%** develop NEOVASC while only **8.7%** develop both.
These percentages are found in Table 2 and 4, respectively.

If you are interested in seeing a breakdown of phenotype presence on the level of patients and eyes, please reference the tables below.

Tables

Eyes that develop NEOVASC and/or HR_TEMPORAL				
0: not Present; 1: Questionable; 2: Present				
		HR_TEMPORAL		
		0	1	2
NEOVASC	0	4535	1039	1266
	1	63	47	120
	2	21	20	174
X^2 test: p ~ 1.061e-138; N = 7285				
Note: These are counts of unique eyes that ever develop phenotype, not counts of eye-visits				

Table 1: Counts of eyes for every combination of HR_TEMPORAL and NEOVASC. A chi-square test was performed on these values to test whether the presence of Neovascularization on Hyper-reflectivity and vice-versa.

Ratio of eyes that develop NEOVASC and/or HR_TEMPORAL

0: not Present; 1: Questionable; 2: Present

		HR_TEMPORAL		
		0	1	2
NEOVASC	0	0.623	0.143	0.174
	1	0.009	0.006	0.016
	2	0.003	0.003	0.024

Ratio develop both: 0.049000 (1,1,+1,2+2,2+2,1)

Ratio develop HR_TEMPORAL only: 0.317000 (0,1+0,2)

Ratio develop NEOVASC only: 0.012000 (1,0+2,0)

Table 2: Every combination of HR_TEMPORAL and NEOVASC as a ratio of total eyes ('N' in Table 1). The percentages of eyes that develop both and each individually were calculated by summing their respective cells in the table.

Patients that develop NEOVASC and/or HR_TEMPORAL

0: not Present; 1: Questionable; 2: Present

		HR_TEMPORAL		
		0	1	2
NEOVASC	0	1869	614	810
	1	36	40	106
	2	14	14	158

X² test: p ~ 9.332e-87; N = 3661

Note: These are counts of unique patients that ever develop phenotype, not counts of patient-visits

Table 3: Counts of patients for every combination of HR_TEMPORAL and NEOVASC. A chi-square test was performed on these values to test whether the presence of Neovascularization on Hyper-reflectivity and vice-versa.

Ratio of patients that develop NEOVASC and/or HR_TEMPORAL

0: not Present; 1: Questionable; 2: Present

		HR_TEMPORAL		
		0	1	2
NEOVASC	0	0.511	0.168	0.221
	1	0.010	0.011	0.029
	2	0.004	0.004	0.043

Ratio develop both: 0.087000 (1,1,+1,2+2,2+2,1)

Ratio develop HR_TEMPORAL only: 0.389000 (0,1+0,2)

Ratio develop NEOVASC only: 0.014000 (1,0+2,0)

Table 4: Every combination of HR_TEMPORAL and NEOVASC as a ratio of total patients ('N' in Table 3). The percentages of patients that develop both and each individually were calculated by summing their respective cells in the table.